Serine, threonine, tyrosine

Which residues the phosphorylations in the dataset sit on.

AT A GLANCE

Where	Phospho → STY Plot
Data level	Phosphosite, as uploaded. Filtering and imputation performed elsewhere do not change it.
Needs	A phospho dataset with a PTM_Collapse_key column. Nothing else is read — not the sample values, not the annotation, not PTM_localization.
What is counted	One count per row of the phosphosite matrix, so one per site identifier. Not per peptide, not per protein, and not per sample.
Controls	Only size, format and appearance. There is no cutoff and no condition selector on this page.
Table	Yes, below the plot, downloadable as CSV.
New	Not covered by any previous manual.

AT PHOSPHOSITE LEVEL

The shared **Localization probability cutoff** selects sites across phosphosite views. At 0 it keeps all sites, including those with no probability; above 0 those unknowns are excluded. This is separate from completeness filtering and does not rerun normalization or imputation. An already collapsed phosphoprotein table has its own recorded collapse cutoff.

1 What the plot answers

The plot answers what kind of phosphoproteome the enrichment produced. Serine, threonine and tyrosine are phosphorylated at very different rates in a cell and enriched at very different efficiencies by different chemistries, so the split between the three bars says more about the sample preparation than about the biology — which is exactly what makes it a useful check. A dataset that was meant to be a phosphotyrosine experiment and comes back looking like a

generic one has failed somewhere upstream of any analysis on this page.

The second bar of information is the fourth bar: Unparsed. It appears only when the site identifier could not be read, and it is a statement about the file format rather than about the chemistry.

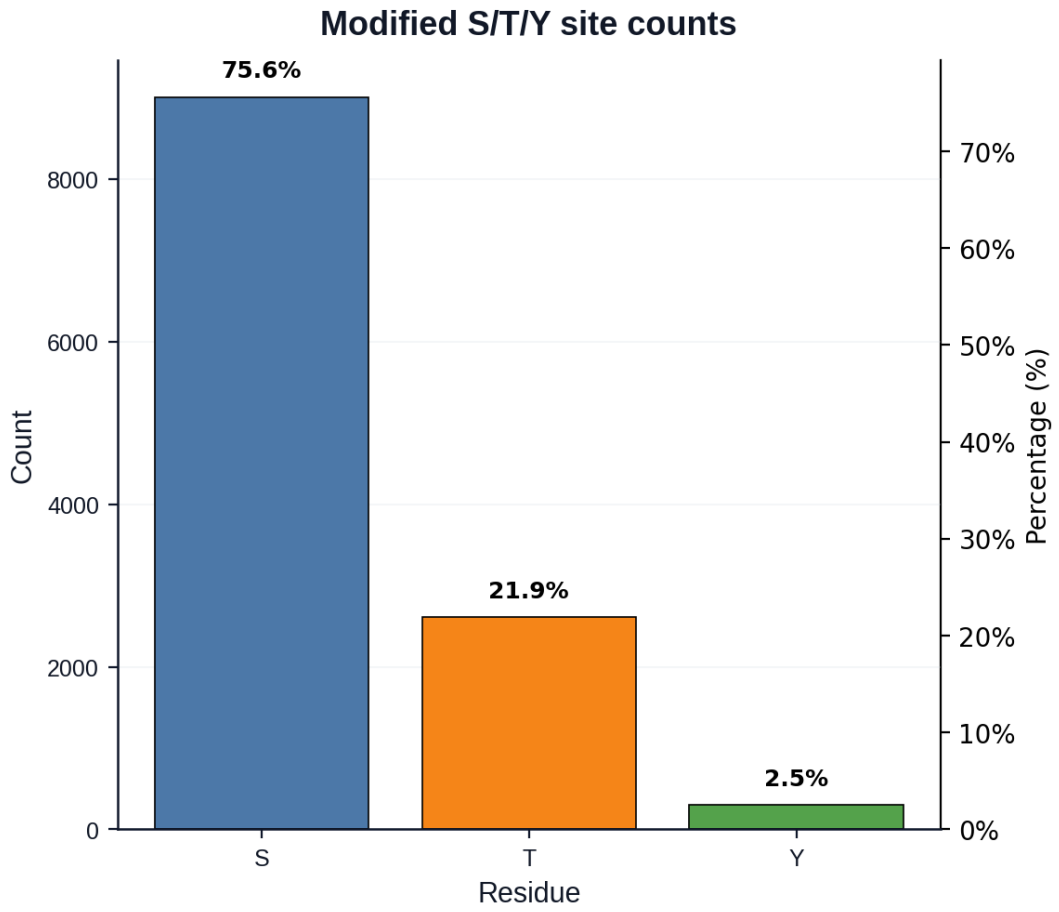


Figure 1.1. The S/T/Y plot on the example dataset, a standard non-selective enrichment: serine 75.6 %, threonine 21.9 % and tyrosine 2.5 % of the rows of the phosphosite matrix. Each bar is a count of site identifiers on the left-hand axis, with its percentage of the total printed above it and the same scale repeated on the right-hand axis. No Unparsed bar appears, so every identifier was read.

Reading it

- One bar per residue, in the fixed order S, T, Y, each in its own colour: blue for serine, orange for threonine, green for tyrosine. A residue that occurs nowhere still gets its bar, at height zero.
- The percentage printed above each bar, and the right-hand axis carrying the same scale. Both are shares of the total number of rows counted, including Unparsed.
- A fourth bar, slate grey and labelled Unparsed, drawn only when at least one identifier could not be read. Its presence means the counts below it are shares of an incompletely parsed dataset.

EASY TO GET WRONG

A high tyrosine share almost always means a phosphotyrosine enrichment worked, not that the biology is unusual. In a cell, phosphotyrosine is a small minority of all phosphorylation; a dataset in which it is not a small minority has been selected for it, by an antibody or by a fractionation, and its serine and threonine bars are then a biased sample of the phosphoproteome rather than a description of it. Read the three bars as a description of the method first and of the sample second.

2 Controls

Controls on this page

Control	What it does	Default	Where
Width (cm), Height (cm), DPI	The size of the *downloaded* figure. The browser scales the rendered figure to fit the page.	17.78 × 11.43, 140	page
Download Plot	Exports the figure in the format chosen in the dropdown beside it: PNG, JPG, WEBP, SVG or PDF.	PNG	page

In the plot appearance panel

Control	What it does	Default	Where
Bar colour	Offered on this page, but the renderer gives each residue its own fixed colour and does not read the setting — see the open question below.	—	appearance
Title, axis labels	Text, size, weight, colour and position, and whether the title is drawn. The plot writes Modified S/T/Y site counts, Residue and Count.	shown	appearance
Font	One face for the whole figure, or one per text.	plot default	appearance
Axis frame, grid, ticks	Which spines are drawn, the grid over them, and the tick labels. The top and right spines are off and a faint horizontal grid is on before the panel touches them.	left and bottom spines, y grid	appearance
Background, theme	Paper colour, transparency, and the preset themes.	default	appearance

The panel shows no **Bars** card on this page: each residue is drawn in its own fixed colour, so a single bar colour would have nothing to apply to.

> **WORTH KNOWING**
>
> The second table is the plot appearance panel, which is hidden until advanced mode is turned on (**Ctrl** + **Shift** + **E**). Everything it contains is described in Part XI; only the entries particular to this plot are repeated here.

3 How it is computed

> **HOW IT IS COMPUTED**
>
> Every row of the phosphosite matrix contributes exactly one count, and the residue is read off its PTM_Collapse_key by two patterns tried in order:
>
> - first ~gene_X###_, which is the shape of an identifier such as A0A590UK10~ARHGEF18_S1035_M1 — a tilde, a gene name, then the residue letter and its position;
> - then, for rows the first did not match, a looser _X### bounded by an underscore or the end of the string, which catches identifiers written without the tilde section.
>
> *X* is S, T or Y in both patterns. A row matched by neither is counted as Unparsed. The counts are then reindexed to the fixed order S, T, Y — so a residue absent from the data still appears at zero — and Unparsed is appended when it is not empty. With N the total of all bars, each bar's label is $100\,c/N$ printed to one decimal, and the right-hand axis is the same conversion applied to the whole y axis.

Edge cases

- Rows are counted, not distinct identifiers. If the phosphosite matrix ever holds the same PTM_Collapse_key twice, that site counts twice.
- Replicates, missing values and intensities never enter this page: it reads one text column. A site counts the same whether it was quantified in every sample or in none.
- PTM_localization is not consulted either. Class I and Not Class I sites are counted alike, so this plot cannot be restricted to confidently localised sites — which matters, because the residue is exactly what a low localisation score is uncertain about.
- A dataset without a PTM_Collapse_key column produces no plot and says so.
- With no rows at all the bars are zero, and the right-hand percentage axis is omitted rather than dividing by zero.

4 The table below the plot

Columns in this table

Column	What it holds
Residue	S, T, Y, and Unparsed when that bar exists.
Count	The number of rows assigned to that residue.
Percentage	Its share of the total of the column above, to two decimals.

The counts always sum to the number of rows in the phosphosite matrix, which is the quickest way to confirm that nothing has been dropped silently.

5 In the notebook

A notebook starting point, using the frames introduced in *The function reference*. These compact examples do not encode every UI option; use the plot's notebook ZIP, where available, for its saved settings and preprocessing recipe:

PYTHON

```
fig = sty_plot(data3)
```

R

```
fig <- sty_plot(data3)
```

The function takes the phosphosite frame and nothing else: the two patterns, the fixed S, T, Y order, the Unparsed bar and the right-hand percentage axis are all as on the page, but there are no size, colour or title arguments, so the figure comes out at the function's own fixed size. Its x axis is labelled Peptide rather than Residue, which is the only wording that differs from the screen.

6 Exporting

The download reproduces the plot at the width, height and resolution set on the page; the figure on screen is fitted to the window instead. PDF for a figure that will be scaled or edited, PNG for a slide. The table downloads separately as CSV.

Serine commonly forms the largest share in broad phosphoproteomic datasets, but there is no required S–T–Y ratio for a valid experiment. Enrichment, search settings, biological material and detection all affect the observed composition. A large tyrosine fraction is compatible with a tyrosine-focused experiment; the bars alone do not establish how it was enriched or which kinase was active. These are counts of reported sites, not phosphorylation occupancies. Inspect Unparsed rows before interpreting percentages, because their residue category is unknown.

Behind this page

Function	What it produces
sty_plot	The bar chart, in the Python package
phospho_sty_table	The three-column table, and the parsing behind it
phospho_sty_png	The figure the page draws